Last Mile Guard: end-to-end system architecture

Google Cloud Run + Vertex AI Gemini · all payments, customers and accounts are made up

Visitor's browser
- React web app
- Live view of each stage

HTTPS

text

Voice input (optional)
- Speech to text in the browser
- Chrome uses Google's service

Customer story
- Made-up payment and note

Google Cloud project last-mile-guard-510814 · region us-central1

Cloud Run service (scales from 0 to 2 copies · 512 MiB · 20 requests per copy)

Static web app
- React pages

FastAPI backend
- /assess /assess/stream /conversation /arena /health

Front-door guards
- Security headers · size limits · rate limits · run caps

What happens to one payment, in order

1. **Validate** — Typed checks on the request
2. **Behaviour rules** — Score from 0 to 100
3. **MuleNet check** — Account network adds points
4. **Message reader** — Gemini spots scam tricks
5. **One question** — Gemini asks and reads the answer
6. **Decision** — Allow, Warn or Hold

Vertex AI calls

Orange = uses Gemini

If Gemini cannot help
- The rules decide alone, and say so

Data and memory
- Made-up data in the image. No database

Vertex AI · Gemini
- Fast model reads and asks
- Smarter model for hard cases
- Location: global

Identity, no keys
- Service account lmg-runtime
- Vertex AI User only

Cloud Logging
- Request logs

new revision

How a change reaches the live site

GitHub
- develop branch

→ **Cloud Shell**
- deploy script and billing check

→ **Cloud Build**
- Builds web pages and server

→ **Artifact Registry**
- Container image

→ **Cloud Run revision**
- Takes all traffic

Offline testing, not part of the live site
- Made-up scenarios replayed from Cloud Shell, scored, results saved to a private repo